

Solutions

Quiz 1

Name: _____

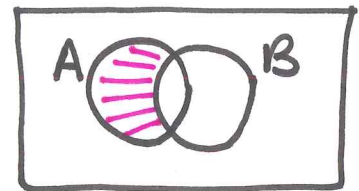
Student ID: _____

Problem 1. Suppose that $P(A) = 0.4$, $P(B) = 0.5$, $P(A \cap B') = 0.1$. Find

(a) $P(A \cup B)$ (b) $P(A \cap B)$ (c) $P(A' \cap B')$.

First solve (b)

The pink highlighted area
is $A \cap B'$, from the graph



$$\begin{aligned} P(A \cap B) &= P(A) - P(A \cap B') \\ &= 0.4 - 0.1 = \underline{\underline{0.3}} \end{aligned}$$

Part (a) $P(A \cup B) = P(A) + P(B) - P(A \cap B)$

$$P(A \cup B) = 0.4 + 0.5 - 0.3 = \underline{\underline{0.6}}$$

Part (c) By De Morgan's law $A' \cap B' = (A \cup B)'$

$$P(A' \cap B') = P((A \cup B)') = 1 - P(A \cup B) = 1 - 0.6 = \underline{\underline{0.4}}$$

Problem 2. Let $P(E)=0.9$ and $P(F)=0.8$. Find the minimum and maximum value for $P(E \cap F)$.

We know $P(E \cup F) \leq 1$.

$$\Rightarrow P(E) + P(F) - P(E \cap F) \leq 1$$

$$\Rightarrow P(E) + P(F) - 1 \leq P(E \cap F)$$

$$\Rightarrow 0.9 + 0.8 - 1 \leq P(E \cap F)$$

$$\Rightarrow 1.7 - 1 \leq P(E \cap F)$$

$$\Rightarrow 0.7 \leq P(E \cap F)$$

$$\Rightarrow \boxed{\text{Min value for } P(E \cap F) \text{ is } 0.7.}$$

The Best scenario will be the case when E or F is the subset of the other. Since

$P(E) > P(F)$, E cannot be a subset of F .

But F could be a subset of E , in that

$$\text{case } F \cap E = F \Rightarrow P(F \cap E) = P(F) = 0.8$$

$$\Rightarrow \boxed{\text{Max value for } P(F \cap E) \text{ is } 0.8}$$

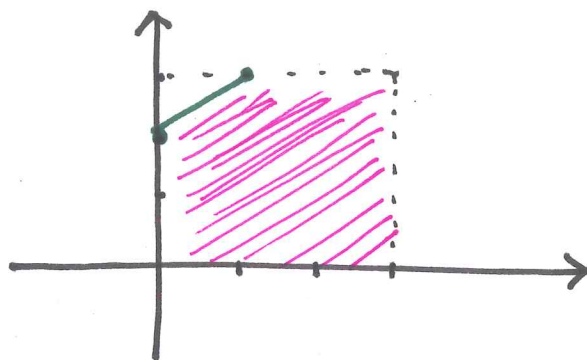
Problem 3. $S = \{(x, y) : 0 \leq x, y \leq 3\}$.

(a) Let $E = \{(x, y) \in S : y - x \leq 2\}$. Find $P(E)$.

(b) Let $F = \{(x, y) \in S : y = x - 1\}$. Find $P(F)$.

The sample space is the square $[0, 3] \times [0, 3]$.

$$\text{Area}(S) = 3 \cdot 3 = 9$$



(a) First find line $y = x + 2$, you can give two points to sketch it for example $(0, 2)$ and $(1, 3)$ are both on the line. The highlighted pink area is E .

$$\begin{aligned} P(E) &= 1 - P(E^c) = 1 - \frac{\text{Area}(E^c)}{\text{Area}(S)} = 1 - \frac{\frac{1 \cdot 1}{2}}{9} \\ &= 1 - \frac{1}{18} = \frac{17}{18} \end{aligned}$$

(b) Area of a line is zero. $P(F) = \frac{0}{9} = 0$.